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Total No. of printed pages = 02

Monsoon, 2025

B.Tech 5th Semester Examination

Plumbing (Water and Sanitation)

Course Code: BCE235PE10T

Full Marks – 50

Time – 2 hours

SECTION 1

Answer all the questions

(Multiple Choice) Marks: 10 x 1 =10

1.

- i. The primary purpose of plumbing codes is to ensure:
 - a) Architectural design aesthetics
 - b) Structural safety only
 - c) Health, safety, and sanitation of users
 - d) Cost reduction
- ii. The *Uniform Illustrated Plumbing Code-India (UIPC-I)* is published by:
 - a) BIS
 - b) IAPMO and IPA
 - c) CPHEEO
 - d) NBC Board
- iii. The term *trap* in plumbing refers to:
 - a) A device to prevent leakage
 - b) A water seal preventing entry of sewer gases
 - c) A component for water storage
 - d) A water treatment device
- iv. The *one-pipe system* in drainage means:
 - a) Separate pipes for soil and waste
 - b) Common pipe for both soil and waste
 - c) Single pipe for stormwater only
 - d) Separate pipes for each fixture
- v. *Rainwater harvesting* includes:
 - a) Use of sub-drains only
 - b) Collection, storage, and reuse of rainwater
 - c) Use of gray water
 - d) Disposal of stormwater
- vi. Potable water means:
 - a) Water suitable for drinking
 - b) Water used for irrigation
 - c) Water for industrial use only
 - d) Water containing minerals
- vii. Gray water refers to:
 - a) Sewage from toilets
 - b) Wastewater from sinks, baths, and washing machines

- c) Industrial effluents
- d) Treated water
- viii. *Pump head* represents:
 - a) Mechanical loss
 - b) Energy imparted by pump to fluid
 - c) Motor efficiency
 - d) Static lift only
- ix. *Aerobic treatment* in STP refers to:
 - a) Treatment without oxygen
 - b) Treatment using oxygen
 - c) Chemical treatment
 - d) Filtration process
- x. *Disinfection in WTP* is mainly done to:
 - a) Remove hardness
 - b) Eliminate pathogens
 - c) Improve colour
 - d) Reduce turbidity

SECTION 2

2. Answer any one question from this section

Marks: 10

- A) Explain the importance of plumbing codes in ensuring health and safety in buildings.
- B) Describe the roles of *UPC-I*, *NBC*, and *Local Municipal Laws* in plumbing design and execution.

SECTION 3

3. Answer any one question from this section

Marks: 10

- A) Define plumbing fixtures and fittings. Explain their installation standards in plan and elevation.
- B) Analyze the impact of hot water temperature control and location of valves on user safety.

SECTION 4

4. Answer any one question from this section

Marks: 10

- A) Explain the purpose of traps and describe different types used in plumbing systems.
- B) Compare the *one-pipe* and *two-pipe* drainage systems with neat sketches.

SECTION 5

5. Answer any one question from this section

Marks: 10

- A) Differentiate *potable*, *non-potable*, *gray*, and *reclaimed* water with examples.
- B) Compare *aerobic* and *anaerobic* sewage treatment methods used in *Sewage Treatment Plants (STPs)*
